

NEXUS

Adversarial Fraud Network Detection

Evidence-driven investigation of coordinated fraud networks under adversarial device and IP rotation.

AI Defense Lab 2026

Track 2 — Fraud, Scam & Identity Defense

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Team Number:

23



Problem & Threat :

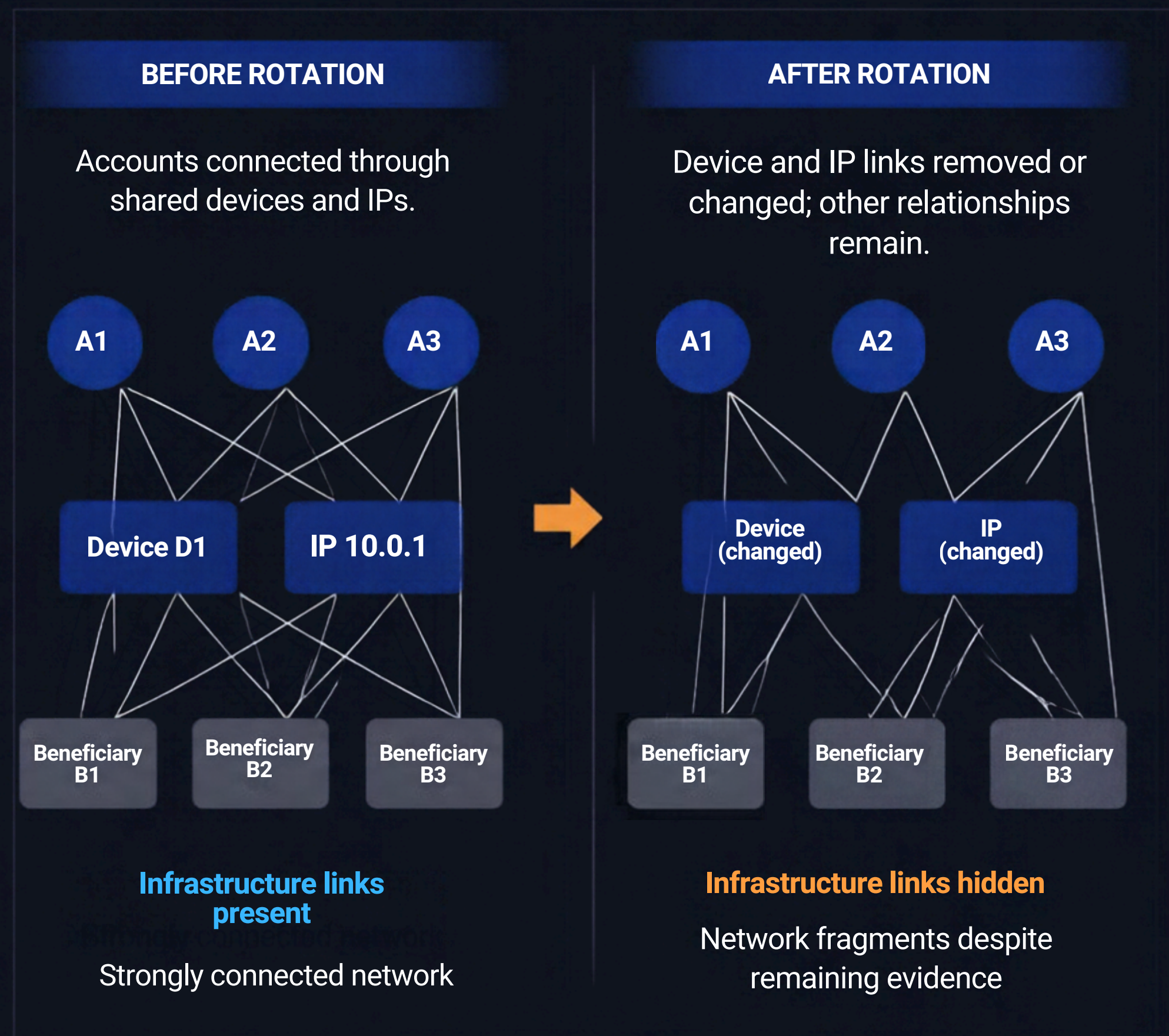
- Coordinated fraud can appear as a network of related accounts, devices, IP addresses, beneficiaries, and behavioral patterns.
- Fraud networks may rotate infrastructure identifiers to weaken infrastructure-based detection.
- If shared **infrastructure disappears**, relationship graphs can fragment even when other evidence remains.

Objective:

Identify coordinated suspicious activity using multiple independent relationship and behavioral signals while minimizing unnecessary friction for legitimate users.

Protected assets :

- User/account trust
- Transaction and identity integrity
- Fraud investigation evidence
- Analyst decision quality



NEXUS: From Signals to Defensible Decisions :



A dashboard alone is not the security solution. NEXUS connects signals to evidence, risk, decision, and auditability.

Candidate-cluster membership is a network-candidate signal, not a final fraud verdict.

High severity is an operational escalation signal, not proof of criminal activity.

System Architecture & Evidence :



EVIDENCE TYPES

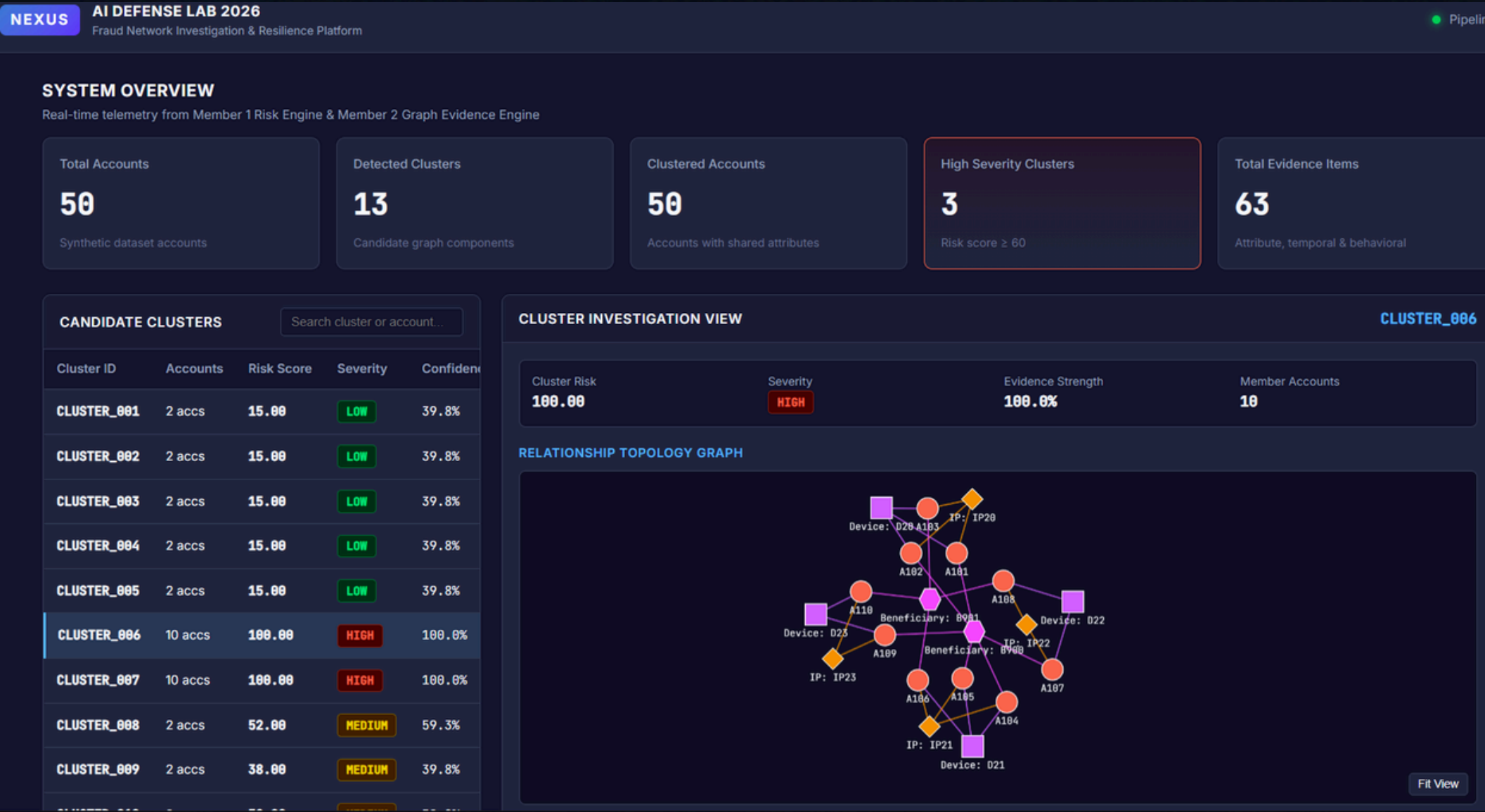
- Shared device
- Shared IP
- Shared beneficiary
- Temporal proximity
- Behavioral overlap

TECHNOLOGIES

- Python
- Pandas
- NetworkX
- Flask
- Gunicorn
- Pytest

SYSTEM BOUNDARIES

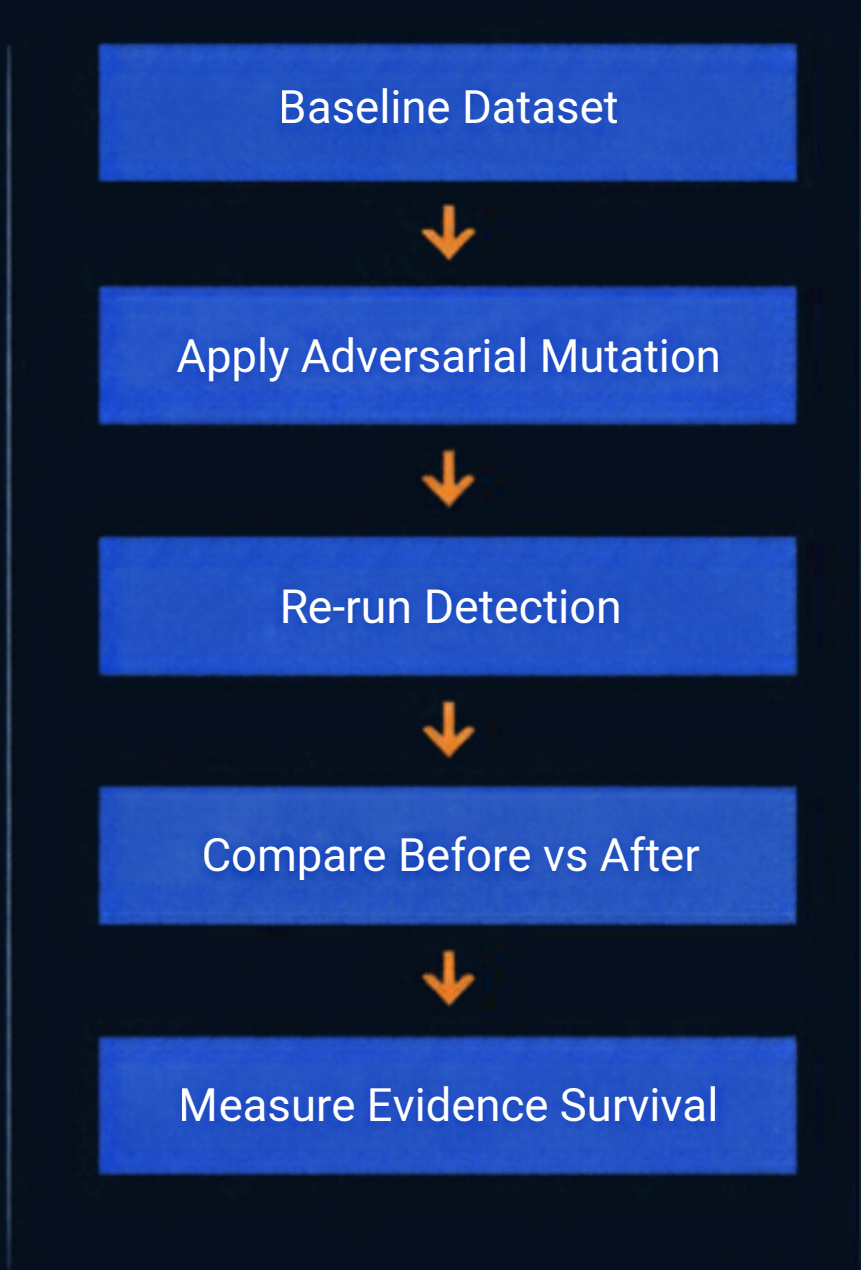
- No real banking API
- No real customer data
- No proprietary fraud API
- No automatic account freezing



Adversarial Resilience: Can the Network Survive Infrastructure Rotation?

Test whether fraud-network evidence remains available when attackers change device and IP infrastructure.

- BASELINE**
No infrastructure mutation
- DEVICE ROTATION**
Shared device relationships are removed
- IP ROTATION**
Shared IP relationships are removed
- COMBINED**
Both shared device and shared IP relationships are removed



RESILIENCE COMPARISON: IP ROTATION			
Metric	Baseline	Attack Scenario	Change (Δ)
Detected Clusters	13	4	-9
Clustered Accounts	50	30	-20
Shared Devices	8	8	0
Shared IPs	23	0	-23
Shared Beneficiaries	6	6	0
Total Evidence Items	63	28	-35
Average Cluster Risk	44.85	88.33	+43.48
Average Confidence	55.7%	73.0%	+0.1731
Evidence Survival Rate	100.0%	44.4%	

Infrastructure rotation fragments the relationship graph, but **residual evidence** from shared beneficiaries, temporal relationships, and behavioral overlap remains.

Clustered accounts decreased from 50 to 30; the 20 accounts no longer clustered were legitimate accounts in the synthetic dataset.

Evaluation & Detection Quality :

The evaluation harness uses synthetic ground truth **AFTER** candidate clusters are generated.

Ground truth does NOT participate in: **graph construction; clustering; scoring; adversarial mutation.**

RESILIENCE COMPARISON								
Clusters	Accounts	Devices	IPs	Beneficiaries	Evidence	Survival	Avg Risk	Avg Confidence
13	50	8	23	6	63	100.0%	44.85	55.7%
13	50	0	23	6	55	87.3%	44.85	53.6%
4	30	8	0	6	28	44.4%	88.33	73.0%
6	30	0	0	6	30	47.6%	92.22	68.7%

MEMBERSHIP METRICS					
Signal	Scenario	Precision	Recall	F1	FPR
Candidate	Baseline	60.0%	100.0%	75.0%	100.0%
High severity	Baseline	100.0%	100.0%	100.0%	0.0%
Candidate	Device Rotation	60.0%	100.0%	75.0%	100.0%
High severity	Device Rotation	100.0%	100.0%	100.0%	0.0%
Candidate	IP Rotation	100.0%	100.0%	100.0%	0.0%
High severity	IP Rotation	100.0%	100.0%	100.0%	0.0%
Candidate	Combined Rotation	100.0%	100.0%	100.0%	0.0%
High severity	Combined Rotation	100.0%	100.0%	100.0%	0.0%

All four cases : Precision 100%; Recall 100%; F1 100%; False-positive Rate 0%.

Synthetic evaluation; not a production benchmark.

Testing, Safety & Human Control :

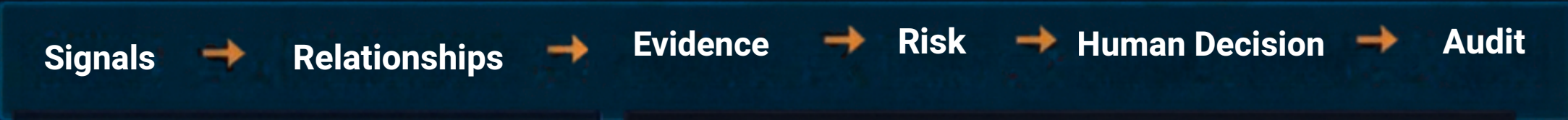
01	NORMAL CASE	Legitimate activity is handled without unnecessary escalation.	HUMAN CONTROL	PRIVACY	AUDITABILITY	
	02	POSITIVE / ATTACK CASE				Coordinated suspicious accounts form clusters and produce evidence.
	03	NEGATIVE / FALSE-POSITIVE CASE				Legitimate shared relationships create candidate clusters requiring analyst review.
	04	FAILURE CASE				System safely handles invalid, insufficient, or unexpected input without uncontrolled actions.
	05	ADVERSARIAL CASE				Device rotation, IP rotation, combined rotation test infrastructure-dependent evidence survival.

Nexus does not automatically freeze accounts.	Synthetic data only.	Each decision records:
Final operational decisions remain with a human analyst.	No confidential personal data.	Cluster ID
Decision states: ALLOW / REVIEW / ESCALATE	No real banking transactions.	Decision
	No real banking transactions.	Reason
		Timestamp

AUDIT LOG EXAMPLES		
Cluster ID	Decision	Reason
CLUSTER 006	ESCALATE	multiple independent relationship and behavioral signals
CLUSTER_013	REVIEW	strong relationship evidence
CLUSTER 001	ALLOW	legitimate shared office network verified

Conclusion: Evidence Before Action :

Nexus treats fraud detection as an investigation problem rather than a simple block/allow decision.



What NEXUS demonstrates

- Multiple relationship and behavioral signal can support fraud network investigation.
- Infrastructure rotation can fragment relationship graphs.
- Residual evidence can survive through other signals.
- Evidence can be exposed alongside risk and confidence.
- Human analysis can make the final decision.
- Adversarial tests can be repeated and compared quantitatively.

Key limitation :

Infrastructure-dependent graph detection is vulnerable to identity / infrastructure rotation.

Combined rotation effect

RESILIENCE COMPARISON: COMBINED			
Metric	Baseline	Attack Scenario	Change (Δ)
Detected Clusters	13	6	-7
Clustered Accounts	50	30	-20
Shared Devices	8	0	-8
Shared IPs	23	0	-23
Shared Beneficiaries	6	6	0
Total Evidence Items	63	30	-33
Average Cluster Risk	44.85	92.22	+47.37
Average Confidence	55.7%	68.7%	+0.1297
Evidence Survival Rate	100.0%	47.6%	

Important interpretation

The 20 accounts no longer clustered were legitimate accounts in the synthetic dataset, showing that graph fragmentation can also affect legitimate relationships.

Nexus relationship graph (synthetic)

